

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION SYSTEMS

Compiler Construction (CS F363)

II Semester 2024-25

Compiler Project

Coding Details

(March 15, 2025)

Group Number

17

1. Team Members Names and IDs

ID	<u>2022A7PS0173P</u>	Name	<u>Ameesh Sethi</u>
ID	<u>2022A7PS0097P</u>	Name	<u>Aviral Gupta</u>
ID	<u>2022A7PS0146P</u>	Name	<u>Yatharth Singh</u>
ID	<u>2022A7PS1188P</u>	Name	<u>Armaan Sethi</u>
ID	<u>2022A7PS0052P</u>	Name	<u>Utkarsh Tiwari</u>

2. Mention the names of the Submitted files :

1 README.md	7 lexer.c	13 parser.c	19 testcase3.txt
2 coding details.pdf	8 lexer.h	14 parser.h	20 testcase4.txt
3 driver.c	9 lexerDef.h	15 parserDef.h	21 testcase5.txt
4 grammar.txt	10 makefile	16 structure.md	22 testcase6.txt
5 lexemesandtokens_t1.txt	11 my.txt	17 testcase1.txt	23 trie.c
6 lexemesandtokens_t2.txt	12 output_testcase1.txt	18 testcase2.txt	24 trie.h
25 utils.c	26 utils.h	27 vector.c	28 vector.h

3. Total number of submitted files (including copy the pdf file of this coding details pro forma) : 28 (All files should be in ONE folder named as Group_#)

4. Have you compressed the folder as specified in the submission guidelines? (yes/no) Yes

5. Lexer Details:

- [A]. Technique used for pattern matching: Maximal-Munch Rule - DFA State/Transitions used to match the patterns while doing explicit checking for identifier lengths.
- [B]. Keyword Handling Technique: Trie
- [C]. Hash function description, if used for keyword handling: Trie used to ensure optimal search time and prevention of collisions.
- [D]. Have you used twin buffer? (Yes/No): Yes
- [E]. Error handling and reporting (Yes/No): Yes
- [F]. Describe the errors handled by you: Unknown Pattern/Symbol Errors, Variable/Function Identifier etc.
- [G]. Data Structure Description for tokenInfo (in maximum two lines): tokenInfo pointer points to struct TokenInfo having fields as list of tokens retrieved from TwinBuffer, associated lexeme, the line number, and counter for parsed tokens.

6. Parser Details:

- [A]. High Level Data Structure Description (in maximum three lines each, avoid giving C definitions used):
- grammar : Array of Linked List of Arrays (3 dimensions - Start Variable i.e. LHS (non - terminals), the Linked List of Derivations, Associated Derivation).
 - FIRST and FOLLOW sets : 2D Array. Non - terminal is the first dim and associated terminals are the second dim.
 - parse table: 2D Array. Non - terminal is the first dim and associated terminals are the second dim.

iv. parse tree: (Describe the node structure also) : **n-D tree. Each node has information about variable, counter for number of child nodes and pointers to them.**

v. Any other (specify and describe) **None**

[B]. Parse tree

- i. Constructed (yes/no): **Yes**
- ii. Printing as per the given format (yes/no): **Yes**
- iii. Describe the order you have adopted for printing the parse tree nodes (in maximum two lines)
Inorder (first, leftmost node, then parent node and then siblings except leftmost node.)

[C]. Grammar and Computation of First and Follow Sets

- i. Data structure for original grammar rules: **Array of Linked List of Arrays**
- ii. FIRST and FOLLOW sets computation automated (Yes/No): **Yes**
- iii. Name the functions (if automated) for computation of First and Follow set :
ComputeFirstAndFollowSets
- iv. If computed First and Follow sets manually and represented in file/function (name that) : **NA**

[D]. Error Handling

- v. Attempted (yes/ no): **Yes**
- vi. Describe the types of errors handled :
 1. **Errors handled include errors like Keyword Spell Check, Missing Operator, Structural Inconsistencies, Unbalanced Parentheses et cetera while preventing infinite loops in the parser.**
 2. **Addresses cases where a terminal on the stack does not match the next input token or when a non-terminal fails to produce a valid derivation**
 3. **Ensures proper termination by handling scenarios where the stack is empty with remaining input tokens or when parsing completes but the stack is not fully consumed.**

7. Compilation Details:

[A]. Makefile works (Yes/No): **Yes**

[B]. Code Compiles (Yes/No): **Yes**

[C]. Mention the .c files that do not compile: **NA**

[D]. Any specific function that does not compile: **NA**

[E]. Ensured the compatibility of your code with the specified gcc version (yes/no) **Yes**

8. Driver Details: Does it take care of the options specified earlier (yes/no): **Yes**

9. Execution

[A]. status (describe in maximum 2 lines): **Compiled and executed correctly. Error recovery has been implemented.**

[B]. Gives segmentation fault with any of the test cases (1-6) uploaded on the course page. If yes, specify the testcase file name: **NA**

10. Specify the language features your lexer or parser is not able to handle (in maximum one line) : **NA**

11. Are you availing the lifeline (Yes/No): **NO**

12. Declaration: We, Ameesh Sethi, Aviral Gupta, Yatharth Singh, Armaan Sethi, Utkarsh Tiwari, declare that we have put our genuine efforts in creating the compiler project code and have submitted the code developed only by us. We have not copied any piece of code from any source. If our code is found plagiarized in any form or degree, we understand that a disciplinary action as per the institute rules will be taken against all of us in our team and we will accept the penalty as decided by the department of Computer Science and Information Systems, BITS, Pilani.

Your names and IDs

Name: Ameesh Sethi

ID: 2022A7PS0173P

Name: Aviral Gupta

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Date: 15/03/2025

Not to exceed 3 pages.